

Using Electrodynamic Tether Technology for Sustainable Satellite Deorbiting and Space Debris Reduction

Executive Summary

As the number of satellites launched into Low Earth Orbit (LEO) continues to grow rapidly, space debris has become one of the most critical challenges threatening the future sustainability of space operations. Defunct satellites, abandoned rocket stages, and fragmented debris continuously increase the risk of orbital collisions, threatening active missions, communication systems, Earth observation services, and future exploration activities.

This project presents an innovative and sustainable solution based on Electrodynamic Tether (EDT) technology. The concept focuses on integrating a conductive tether system into satellites so that, after mission completion, the tether deploys automatically and interacts with Earth's magnetic field to generate a Lorentz force. This force acts as an electrodynamic drag that accelerates the deorbiting process without requiring large amounts of chemical propellant.

The project combines aerospace engineering, orbital mechanics, simulation technologies, and Artificial Intelligence to create a smart and efficient deorbiting system for future satellite missions. MATLAB and STK were used to simulate orbital behavior and tether performance, while Power BI was utilized for visualization and comparative analysis. In addition, AI-based predictive models were developed to estimate deorbit time and forecast future debris growth trends.

By reducing orbital lifetime and enabling autonomous end-of-life disposal, the proposed system contributes directly to sustainable space operations and long-term orbital safety.

The Space Debris Problem

Space debris has become a major global concern in the modern space era. Thousands of inactive satellites and millions of debris fragments currently orbit Earth at extremely high velocities. Even very small debris particles can cause catastrophic damage when colliding with operational spacecraft.

The rapid expansion of satellite constellations and commercial space activities increases orbital congestion and raises the probability of chain-reaction collisions, commonly known as the Kessler Syndrome. Such a scenario could severely affect future access to space and compromise critical technologies that modern societies depend on, including navigation systems, weather forecasting, communications, scientific research, and disaster monitoring.

Traditional satellite deorbiting methods often rely on onboard propulsion systems and remaining fuel reserves. However, many satellites reach the end of their operational life without sufficient propellant to safely deorbit. This creates long-term orbital pollution and increases environmental risks in space.

Therefore, the development of autonomous, low-cost, and fuel-efficient deorbiting technologies is essential for ensuring the sustainability of future space missions.

Proposed Electrodynamic Tether Solution

The proposed solution is based on integrating an Electrodynamic Tether system into future satellites operating in Low Earth Orbit.

An Electrodynamic Tether is a long conductive tape or wire deployed from the satellite after the mission ends. As the satellite moves through Earth's magnetic field, an electromotive force is naturally induced along the tether. This interaction generates an electrical current, and the resulting interaction between the current and the geomagnetic field produces a Lorentz force.

In deorbiting mode, this Lorentz force acts opposite to the satellite's orbital motion, gradually reducing orbital energy and lowering altitude until atmospheric re-entry occurs.

One of the major advantages of EDT technology is that it offers a propellant-less or low-fuel deorbiting solution. Instead of depending entirely on chemical propulsion systems, the tether uses naturally available environmental conditions in space, making the process more sustainable and economically efficient.

Compared to conventional deorbiting techniques, Electrodynamic Tethers provide several important advantages:

- Reduced dependence on onboard fuel
- Autonomous end-of-life disposal
- Lower mission cost
- Improved orbital sustainability
- Reduced long-term debris accumulation
- Compatibility with small satellites and CubeSats

Simulation and AI Integration

To validate the feasibility of the proposed concept, multiple engineering and simulation tools were integrated into the project.

MATLAB was used for mathematical modeling and simulation of orbital decay behavior, Lorentz force interaction, and tether-related calculations. STK was utilized to model orbital trajectories and compare natural orbital decay with tether-assisted deorbiting scenarios.

Power BI was used to visualize mission data, orbital trends, and comparative analyses in a clear and interactive way.

In addition to simulation tools, Artificial Intelligence played an important role in the project. AI-based prediction models were developed to predict satellite deorbit time, analyze orbital decay behavior, forecast future growth of space debris, and support future autonomous orbital management systems.

Sustainability and Environmental Impact

The proposed Electrodynamic Tether system directly supports sustainable space operations by enabling satellites to safely remove themselves from orbit after completing their missions. This reduces the number of inactive objects remaining in space and decreases future collision risks.

Unlike traditional propulsion systems that consume large amounts of fuel, EDT systems rely primarily on electromagnetic interaction with Earth's magnetic field. This makes the technology more environmentally friendly and more suitable for future large-scale satellite operations.

Future Vision and Applications

Electrodynamic Tether technology has the potential to become an important component of future satellite missions and orbital infrastructure.

In the future, EDT systems could be integrated into commercial satellite constellations, CubeSat missions, Earth observation satellites, scientific spacecraft, and autonomous debris-removal missions.

Conclusion

This project presents a sustainable and innovative approach to addressing one of the most urgent challenges in modern space activities: space debris accumulation.

By combining Electrodynamic Tether technology with simulation tools, orbital analysis, and Artificial Intelligence, the project offers a practical and scalable solution for autonomous satellite deorbiting and orbital sustainability.

As humanity enters a new era of intensive space utilization, sustainable orbital management will play a critical role in protecting the future of space exploration and global satellite infrastructure.